

STAT 423 Winter 2026 - Final Project

Krish Doshi, Lauren Yee, Kian Sharifzadeh

2026-03-09

Data Processing - Krish Doshi

```
orig_data <- read.csv("billboard.csv")

cleaned_data <- data.frame(artist = orig_data$artist, rating = orig_data$overall_rating,
  age=orig_data$front_person_age)

cleaned_data$gender <- ifelse(orig_data$artist_male == 1, "Male", "Not Male")

cleaned_data$race <- ifelse(orig_data$artist_white == 1, "White", "Not White")

cleaned_data$genre <- ifelse(is.na(orig_data$cdr_genre), "Genre Not Specified",
  orig_data$cdr_genre)

cleaned_data$genre <- ifelse(grepl("Pop", orig_data$cdr_genre), "Pop",
  ifelse(grepl("Country", orig_data$cdr_genre), "Country",
  ifelse(grepl("Hip-Hop", orig_data$cdr_genre), "Hip-Hop",
  ifelse(grepl("Rock", orig_data$cdr_genre), "Rock",
  ifelse(grepl("Jazz", orig_data$cdr_genre), "Jazz", "Other")))))

cleaned_data <- cbind(cleaned_data, bpm=orig_data$bpm, energy=orig_data$energy,
  danceability = orig_data$danceability, happiness = orig_data$happiness,
  loudness_db = orig_data$loudness_d_b,
  length_sec = orig_data$length_sec)

cleaned_data$year <- as.numeric(substr(orig_data$date, 1,
  unlist(gregexpr("-", orig_data$date))[1] - 1))

cleaned_data <- cleaned_data[rowSums(is.na(cleaned_data)) == 0 , ]

model_results <- list()

for(i in seq(1960, 2030, 10)){
  decade_label <- paste0(i - 10, "s")

  subpop <- cleaned_data[cleaned_data$year >= i - 10 &
    cleaned_data$year < i, ]

  # --- SAFETY CHECK 1: Ensure enough rows exist ---
  # If you have < 50 rows, the interaction model will likely fail.
```

```

if(nrow(subpop) < 50) {
  message(paste("Skipping", decade_label, "- Not enough data points."))
  next
}

factor_counts <- sapply(subpop[, c("genre", "gender", "race")],
  function(x) length(unique(na.omit(x))))

if(any(factor_counts < 2)) {
  message(paste("Skipping", decade_label, "due to single-level factor:",
    names(factor_counts)[factor_counts < 2]))
  next
}

first_model <- lm(data = subpop, rating ~ age + race + gender + genre + energy
  + danceability + happiness + bpm + loudness_db + length_sec)

second_model <- lm(data = subpop, rating ~ age + gender + race + age*genre
  + age*energy + age*danceability + age*happiness + age*bpm
  + age*loudness_db + age*length_sec + race*genre + race*energy
  + race*danceability + race*happiness + race*bpm + race*loudness_db
  + race*length_sec + gender*genre + gender*energy + gender*danceability
  + gender*happiness + gender*bpm + gender*loudness_db + gender*length_sec)

if(anova(first_model, second_model)$"Pr(>F)"[2] <= 0.1){
  optimal_model <- step(second_model, direction="backward")
} else {
  optimal_model <- step(first_model, direction="backward")
}

model_results[[decade_label]] <- list(
  "first" = first_model,
  "second" = second_model,
  "optimal" = optimal_model
)

print(decade_label)

summary(first_model)

summary(second_model)

summary(optimal_model)
}

```

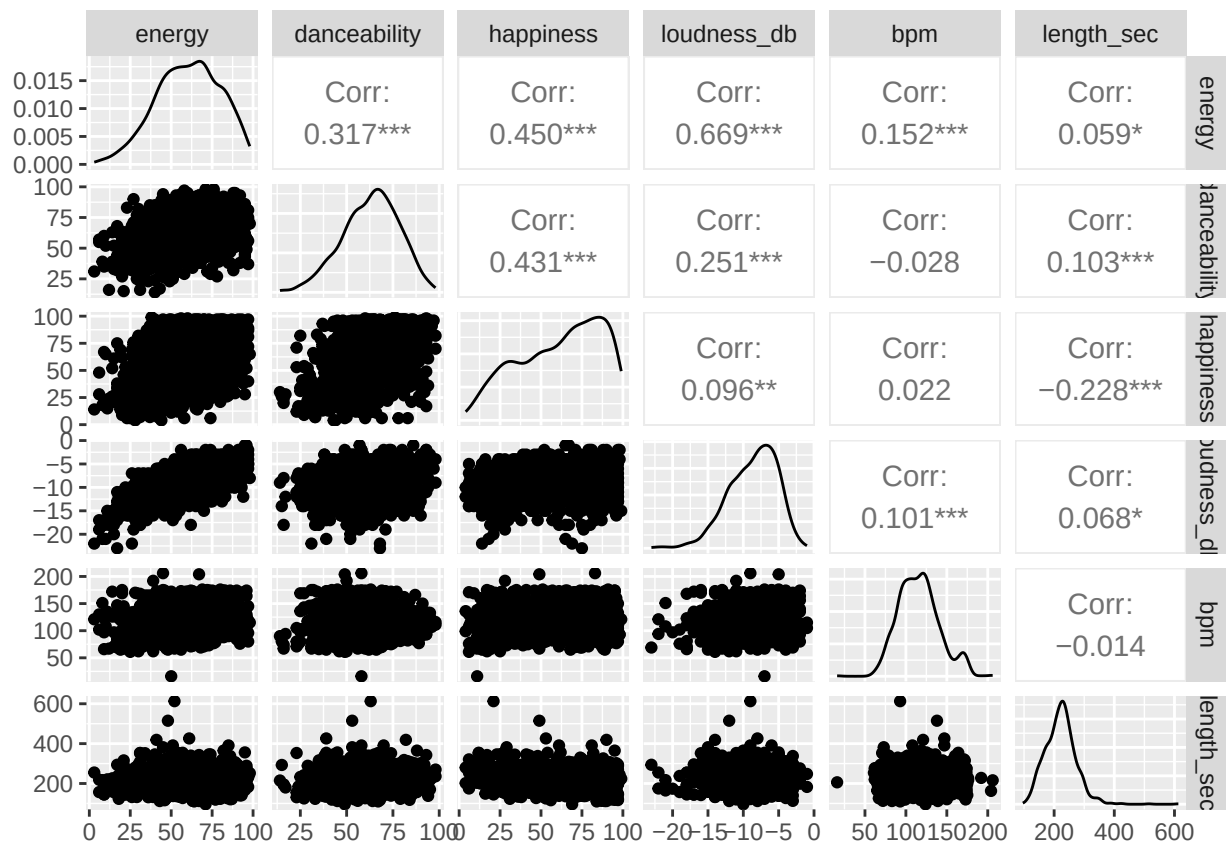
```
## Skipping 1950s - Not enough data points.
```

```
## Skipping 2020s due to single-level factor: genre
```

```

library("GGally")
ggpairs(data = cleaned_data[,c("energy", "danceability", "happiness", "loudness_db", "bpm", "length_sec

```



Loop through each decade in the model_results list

```
for (decade in names(model_results)) {
```

Check if the 'optimal' model exists for this decade

```
if (!is.null(model_results[[decade]]$optimal)) {
```

```
  cat("\n", "=====", "\n")
```

```
  cat(" SUMMARY FOR OPTIMAL MODEL:", decade, "\n")
```

```
  cat("=====", "\n")
```

Print the summary

```
  print(summary(model_results[[decade]]$optimal))
```

```
} else {
```

```
  cat("\nNo optimal model found for:", decade, "\n")
```

```
}
```

```
}
```

```
##
```

```
## =====
```

```
## SUMMARY FOR OPTIMAL MODEL: 1960s
```

```
## =====
```

```
##
```

```
## Call:
```

```
## lm(formula = rating ~ age + gender + race + genre + energy +
```

```
## danceability + happiness + length_sec + race:genre + race:energy +
```

```
## race:danceability + gender:danceability + gender:happiness,
```

```
## data = subpop)
```

```
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -5.3064 -1.2960  0.1313  1.2436  4.2045
##
## Coefficients: (1 not defined because of singularities)
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    6.235892   2.001360   3.116  0.00214 **
## age           -0.036813   0.021122  -1.743  0.08309 .
## genderNot Male -1.820687   1.293523  -1.408  0.16101
## raceWhite      1.604179   1.795663   0.893  0.37287
## genreJazz      1.473039   1.332354   1.106  0.27040
## genreOther     1.978211   0.999147   1.980  0.04926 *
## genrePop       0.988531   0.889417   1.111  0.26788
## genreRock      0.260014   0.629236   0.413  0.67994
## energy        -0.005765   0.015336  -0.376  0.70743
## danceability   -0.013030   0.022465  -0.580  0.56264
## happiness      0.004845   0.009013   0.538  0.59154
## length_sec     0.008965   0.003791   2.365  0.01912 *
## raceWhite:genreJazz -1.353373  2.222521  -0.609  0.54334
## raceWhite:genreOther -3.249204  1.187123  -2.737  0.00683 **
## raceWhite:genrePop -1.786329  0.784348  -2.277  0.02395 *
## raceWhite:genreRock      NA         NA      NA      NA
## raceWhite:energy    0.024399  0.017347   1.407  0.16132
## raceWhite:danceability -0.046982  0.022464  -2.091  0.03791 *
## genderNot Male:danceability 0.076584  0.023552   3.252  0.00137 **
## genderNot Male:happiness -0.030694  0.016106  -1.906  0.05830 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.797 on 178 degrees of freedom
## Multiple R-squared:  0.314, Adjusted R-squared:  0.2446
## F-statistic: 4.527 on 18 and 178 DF, p-value: 4.551e-08
##
##
## =====
## SUMMARY FOR OPTIMAL MODEL: 1970s
## =====
##
## Call:
## lm(formula = rating ~ age + race + genre + length_sec, data = subpop)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -5.1615 -1.0209  0.0511  1.0964  5.0519
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    5.250133   0.992268   5.291 2.77e-07 ***
## age           -0.033100   0.020798  -1.591  0.112830
## raceWhite     -0.430183   0.299921  -1.434  0.152800
## genreJazz     -0.547224   1.924576  -0.284  0.776402
## genreOther     0.979427   0.660001   1.484  0.139144
## genrePop      -0.607790   0.641502  -0.947  0.344375
```

```

## genreRock    0.182210    0.658709    0.277 0.782315
## length_sec   0.009494    0.002515    3.775 0.000202 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.808 on 237 degrees of freedom
## Multiple R-squared:  0.2243, Adjusted R-squared:  0.2013
## F-statistic: 9.787 on 7 and 237 DF,  p-value: 1.045e-10
##
##
## =====
## SUMMARY FOR OPTIMAL MODEL: 1980s
## =====
##
## Call:
## lm(formula = rating ~ energy + danceability + happiness, data = subpop)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -3.2377 -1.0513 -0.0178  0.9701  4.2309
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   4.960976    0.549218   9.033  <2e-16 ***
## energy         0.009412    0.005806   1.621   0.1064
## danceability   0.015115    0.009255   1.633   0.1038
## happiness     -0.011560    0.005495  -2.104   0.0365 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.505 on 224 degrees of freedom
## Multiple R-squared:  0.02279, Adjusted R-squared:  0.009699
## F-statistic: 1.741 on 3 and 224 DF,  p-value: 0.1594
##
##
## =====
## SUMMARY FOR OPTIMAL MODEL: 1990s
## =====
##
## Call:
## lm(formula = rating ~ race + gender + energy + happiness + loudness_db,
##     data = subpop)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -3.2837 -0.9790 -0.0504  0.8700  4.4145
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   9.465767    0.942926  10.039  < 2e-16 ***
## raceWhite     -0.469197    0.277569  -1.690   0.09337 .
## genderNot Male  0.720941    0.253791   2.841   0.00523 **
## energy        -0.035118    0.010721  -3.276   0.00135 **
## happiness      0.016936    0.006634   2.553   0.01185 *

```

```

## loudness_db      0.311544    0.066407    4.691 6.82e-06 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.469 on 129 degrees of freedom
## Multiple R-squared:  0.2445, Adjusted R-squared:  0.2152
## F-statistic: 8.349 on 5 and 129 DF,  p-value: 7.287e-07
##
##
## =====
## SUMMARY FOR OPTIMAL MODEL: 2000s
## =====
##
## Call:
## lm(formula = rating ~ age + gender + race + genre + energy +
##     danceability + happiness + bpm + length_sec + age:danceability +
##     race:genre + race:danceability + race:happiness + race:bpm +
##     race:length_sec + gender:danceability + gender:happiness +
##     gender:bpm, data = subpop)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.80175 -0.88847 -0.00231  1.02953  2.93881
##
## Coefficients: (2 not defined because of singularities)
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    14.254799    5.689019   2.506 0.013714 *
## age            -0.262392    0.164223  -1.598 0.113013
## genderNot Male  -0.551642    2.208098  -0.250 0.803196
## raceWhite      -13.479191    3.724145  -3.619 0.000451 ***
## genreOther      0.362481    1.668695   0.217 0.828443
## genrePop        0.218854    1.608297   0.136 0.892013
## genreRock       0.090807    1.604975   0.057 0.954986
## energy          0.018882    0.011341   1.665 0.098826 .
## danceability    -0.083852    0.065397  -1.282 0.202519
## happiness        0.001354    0.011393   0.119 0.905583
## bpm            -0.014549    0.008120  -1.792 0.075985 .
## length_sec      -0.009048    0.005130  -1.764 0.080619 .
## age:danceability  0.003737    0.002195   1.703 0.091492 .
## raceWhite:genreOther -1.186109    0.866090  -1.369 0.173685
## raceWhite:genrePop      NA         NA      NA      NA
## raceWhite:genreRock     NA         NA      NA      NA
## raceWhite:danceability  0.084078    0.031491   2.670 0.008759 **
## raceWhite:happiness    -0.025028    0.018899  -1.324 0.188199
## raceWhite:bpm          0.035295    0.012426   2.840 0.005385 **
## raceWhite:length_sec    0.019703    0.011520   1.710 0.090064 .
## genderNot Male:danceability -0.041970    0.024007  -1.748 0.083262 .
## genderNot Male:happiness  0.021226    0.014487   1.465 0.145771
## genderNot Male:bpm       0.019019    0.010619   1.791 0.076076 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.39 on 108 degrees of freedom
## Multiple R-squared:  0.3417, Adjusted R-squared:  0.2198

```

```
## F-statistic: 2.803 on 20 and 108 DF, p-value: 0.0003367
##
##
## =====
## SUMMARY FOR OPTIMAL MODEL: 2010s
## =====
##
## Call:
## lm(formula = rating ~ gender + energy + happiness + length_sec,
##     data = subpop)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -4.2684 -1.0215  0.1378  0.9163  3.4440
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    4.788777    1.188134   4.031 0.000103 ***
## genderNot Male    0.459667    0.298784   1.538 0.126808
## energy         -0.024064    0.009966  -2.415 0.017405 *
## happiness         0.012193    0.007833   1.557 0.122440
## length_sec       0.009668    0.004602   2.101 0.037952 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.594 on 110 degrees of freedom
## Multiple R-squared:  0.09648, Adjusted R-squared:  0.06363
## F-statistic: 2.937 on 4 and 110 DF, p-value: 0.02378
```

Preliminary Exploration of Data through Visual Analysis

Plots of Individual Relationships

The plots show the relationship between the predictors and the response variable, rating.

```
library(ggplot2)
library(patchwork)

##
## Attaching package: 'patchwork'
##
## The following object is masked from 'package:MASS':
##
##      area

plot_age <- ggplot(cleaned_data, aes(x = age, y = rating)) +
  geom_point(color = "black", size = 1.5) +
  geom_smooth(method = "lm", color = "red", se = F) +
  theme_classic() +
  ggtitle("Rating vs Age")

plot_energy <- ggplot(cleaned_data, aes(x = energy, y = rating)) +
  geom_point(color = "black", size = 1.5) +
  geom_smooth(method = "lm", color = "red", se = F) +
  theme_classic() +
  ggtitle("Rating vs Energy")
```

```
plot_danceability <- ggplot(cleaned_data, aes(x = danceability, y = rating)) +
  geom_point(color = "black", size = 1.5) +
  geom_smooth(method = "lm", color = "red", se = F) +
  theme_classic() +
  ggtitle("Rating vs Danceability")
```

```
plot_happiness <- ggplot(cleaned_data, aes(x = happiness, y = rating)) +
  geom_point(color = "black", size = 1.5) +
  geom_smooth(method = "lm", color = "red", se = F) +
  theme_classic() +
  ggtitle("Rating vs Happiness")
```

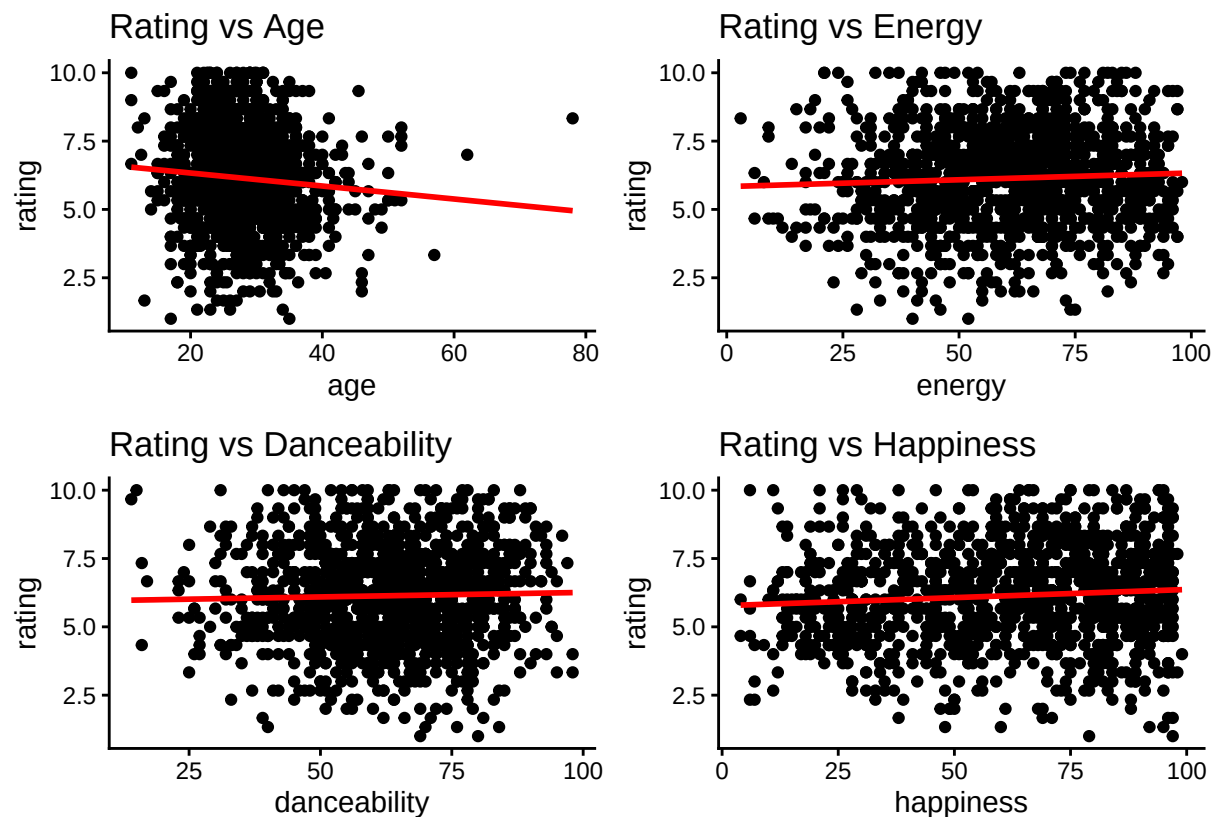
```
plot_age + plot_energy + plot_danceability + plot_happiness
```

```
## `geom_smooth()` using formula = 'y ~ x'
```

```
## `geom_smooth()` using formula = 'y ~ x'
```

```
## `geom_smooth()` using formula = 'y ~ x'
```

```
## `geom_smooth()` using formula = 'y ~ x'
```



```
plot_bpm <- ggplot(cleaned_data, aes(x = bpm, y = rating)) +
  geom_point(color = "black", size = 1.5) +
  geom_smooth(method = "lm", color = "red", se = F) +
  theme_classic() +
  ggtitle("Rating vs BPM")
```

```
plot_loudness_db <- ggplot(cleaned_data, aes(x = loudness_db, y = rating)) +
  geom_point(color = "black", size = 1.5) +
```

```

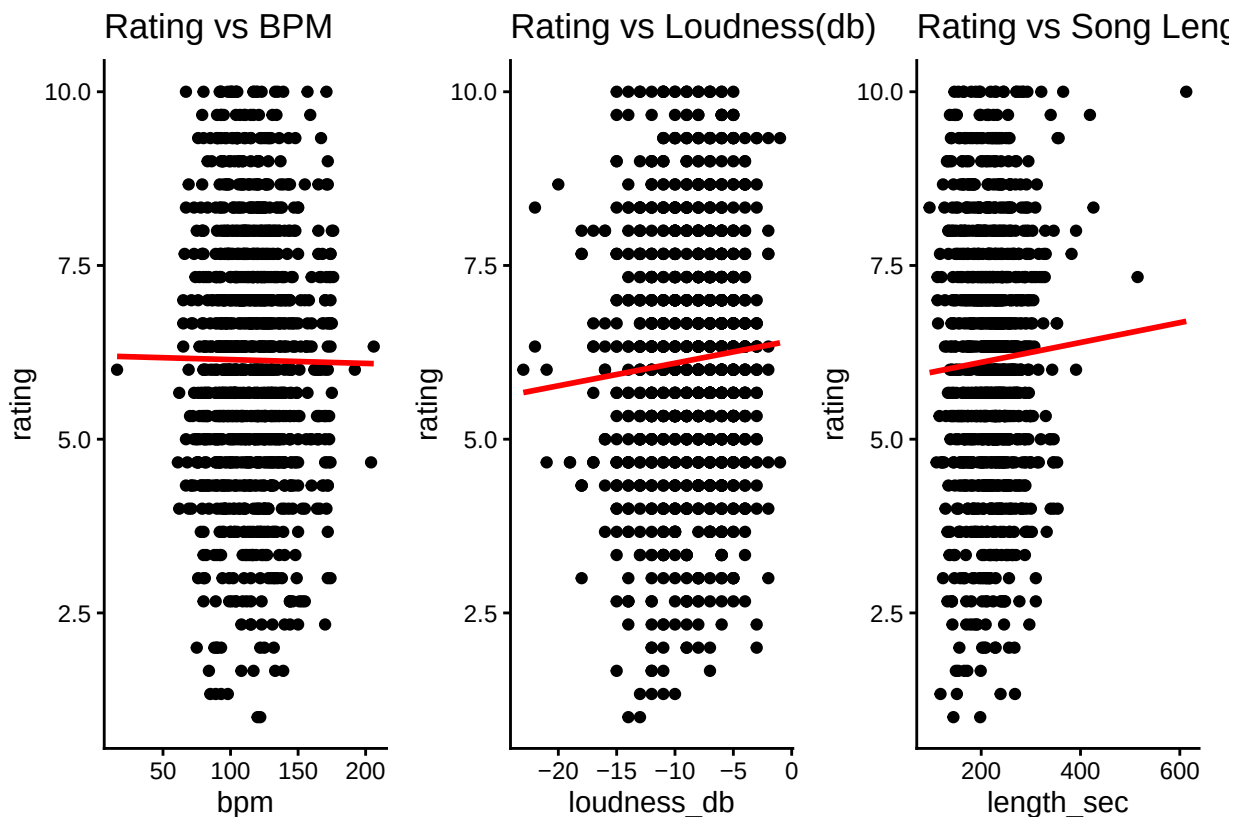
geom_smooth(method = "lm", color = "red", se = F) +
theme_classic() +
ggtitle("Rating vs Loudness(db)")

plot_length_sec <- ggplot(cleaned_data, aes(x = length_sec, y = rating)) +
  geom_point(color = "black", size = 1.5) +
  geom_smooth(method = "lm", color = "red", se = F) +
  theme_classic() +
  ggtitle("Rating vs Song Length(sec)")

plot_bpm + plot_loudness_db + plot_length_sec

## `geom_smooth()` using formula = 'y ~ x'
## `geom_smooth()` using formula = 'y ~ x'
## `geom_smooth()` using formula = 'y ~ x'

```



```

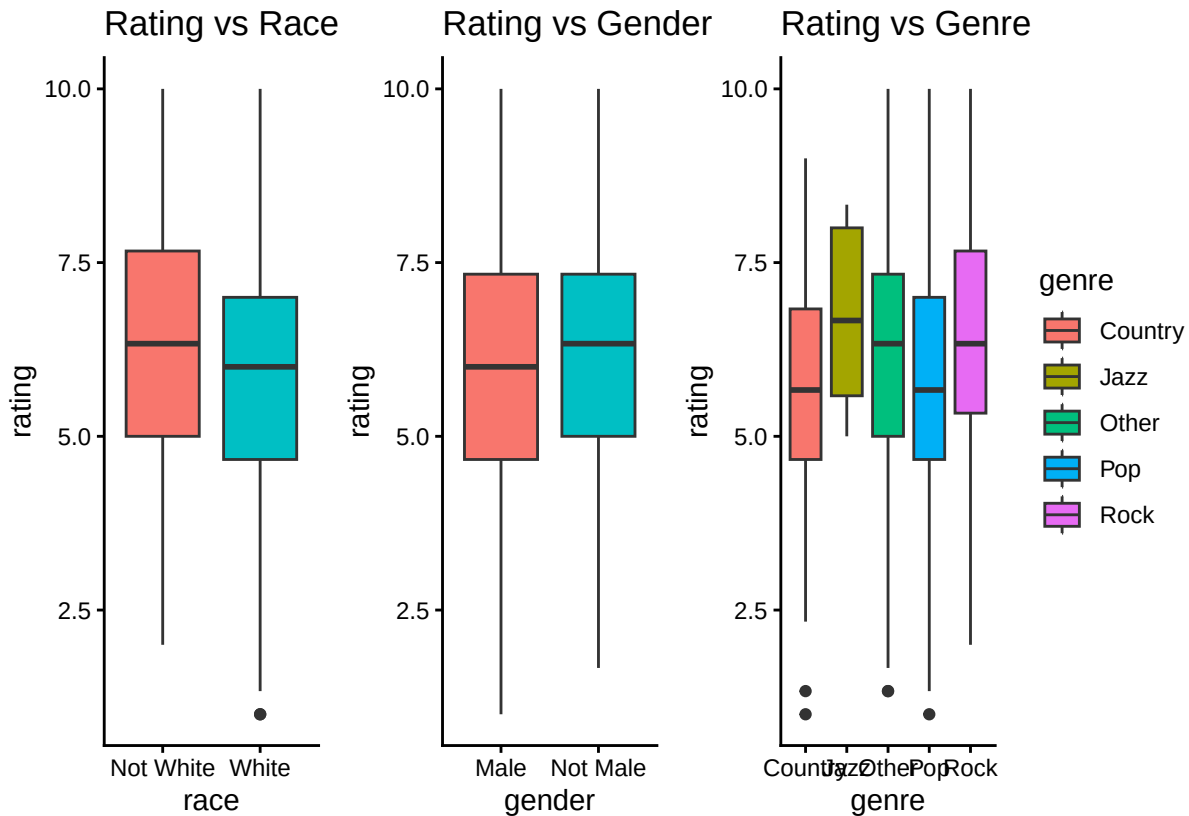
plot_race <- ggplot(cleaned_data, aes(x = race, y = rating, fill = race)) +
  geom_boxplot() +
  theme_classic() +
  ggtitle("Rating vs Race") +
  guides(fill = "none")

plot_gender <- ggplot(cleaned_data, aes(x = gender, y = rating, fill = gender)) +
  geom_boxplot() +
  theme_classic() +
  ggtitle("Rating vs Gender") +
  guides(fill = "none")

```

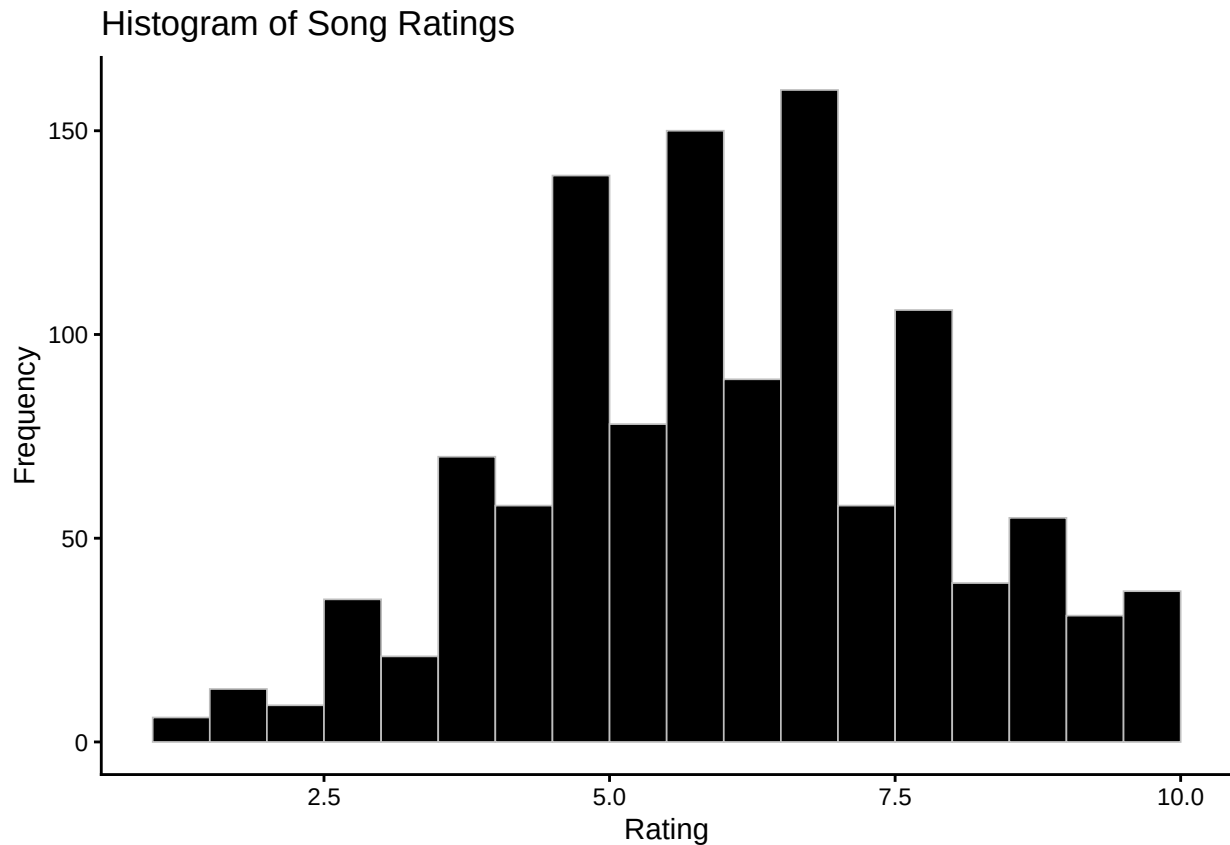
```
plot_genre <- ggplot(cleaned_data, aes(x = genre, y = rating, fill = genre)) +
  geom_boxplot() +
  theme_classic() +
  ggtitle("Rating vs Genre")
```

```
plot_race + plot_gender + plot_genre
```



As expected, banding is observed in the scatterplots because the response variable, rating, is discrete. Since race, gender, and genre are categorical predictors, the relationship with rating is displayed by boxplots instead.

```
ggplot(cleaned_data, aes(x = rating)) +
  geom_histogram(
    binwidth = 0.5,
    boundary = 0.5,
    fill = "black",
    color = "grey",
    linewidth = 0.3
  ) +
  labs(
    title = "Histogram of Song Ratings",
    x = "Rating",
    y = "Frequency"
  ) +
  theme_classic()
```



However, the histogram of ratings appears approximately symmetric and unimodal, resembling a normal distribution. While the ratings are discrete (integer-valued) and there are many categorical predictors, the distribution spans multiple levels, creating a shape that approximates continuity. This supports the treating of rating as a continuous response variable in the regression analysis.

Plots of Grouped Relationships

Data visualizations for grouped relationships help us examine whether relationships between predictors and ratings differ across categories. These plots justify the inclusion of categorical predictors in the regression models and help identify potential interaction effects.

```
model_60s <- model_results[["1960s"]]$optimal
model_70s <- model_results[["1970s"]]$optimal
model_80s <- model_results[["1980s"]]$optimal
model_90s <- model_results[["1990s"]]$optimal
model_2000s <- model_results[["2000s"]]$optimal
model_2010s <- model_results[["2010s"]]$optimal
```

```
summary(model_60s)$r.squared
```

```
## [1] 0.3140162
```

```
summary(model_60s)$adj.r.squared
```

```
## [1] 0.244647
```

```
summary(model_70s)$r.squared
```

```
## [1] 0.2242527
```

```

summary(model_70s)$adj.r.squared

## [1] 0.2013403
summary(model_80s)$r.squared

## [1] 0.02278648
summary(model_80s)$adj.r.squared

## [1] 0.009698796
summary(model_90s)$r.squared

## [1] 0.2444873
summary(model_90s)$adj.r.squared

## [1] 0.2152039
summary(model_2000s)$r.squared

## [1] 0.3416655
summary(model_2000s)$adj.r.squared

## [1] 0.2197517
summary(model_2010s)$r.squared

## [1] 0.09648348
summary(model_2010s)$adj.r.squared

## [1] 0.06362834

vars_by_decade <- lapply(model_results, function(x) {
  attr(terms(x$optimal), "term.labels")
})

all_vars <- unique(unlist(vars_by_decade))

presence_matrix <- sapply(vars_by_decade, function(vars) {
  all_vars %in% vars
})

rownames(presence_matrix) <- all_vars
presence_matrix

##           1960s 1970s 1980s 1990s 2000s 2010s
## age          TRUE  TRUE FALSE  FALSE  TRUE  FALSE
## gender       TRUE FALSE FALSE  TRUE  TRUE  TRUE
## race         TRUE  TRUE FALSE  TRUE  TRUE  FALSE
## genre        TRUE  TRUE FALSE  FALSE  TRUE  FALSE
## energy       TRUE FALSE  TRUE  TRUE  TRUE  TRUE
## danceability TRUE FALSE  TRUE  FALSE  TRUE  FALSE
## happiness    TRUE FALSE  TRUE  TRUE  TRUE  TRUE
## length_sec   TRUE  TRUE FALSE  FALSE  TRUE  TRUE
## race:genre   TRUE FALSE  FALSE  FALSE  TRUE  FALSE
## race:energy  TRUE FALSE  FALSE  FALSE  FALSE  FALSE

```

```
## race:danceability    TRUE FALSE FALSE FALSE    TRUE FALSE
## gender:danceability  TRUE FALSE FALSE FALSE    TRUE FALSE
## gender:happiness    TRUE FALSE FALSE FALSE    TRUE FALSE
## loudness_db         FALSE FALSE FALSE    TRUE FALSE FALSE
## bpm                 FALSE FALSE FALSE FALSE    TRUE FALSE
## age:danceability    FALSE FALSE FALSE FALSE    TRUE FALSE
## race:happiness      FALSE FALSE FALSE FALSE    TRUE FALSE
## race:bpm            FALSE FALSE FALSE FALSE    TRUE FALSE
## race:length_sec     FALSE FALSE FALSE FALSE    TRUE FALSE
## gender:bpm          FALSE FALSE FALSE FALSE    TRUE FALSE
```

```
rownames(presence_matrix)[rowSums(presence_matrix) == ncol(presence_matrix)]
```

```
## character(0)
```

```
presence_df <- as.data.frame(as.table(presence_matrix))
```

```
ggplot(presence_df, aes(x = Var2, y = Var1, fill = Freq)) +
  geom_tile() +
  scale_fill_manual(values = c("black", "#23DC6D")) +
  labs(x = "Decade", y = "Variable", fill = "Included") +
  theme_minimal()
```



```
true_counts <- rowSums(presence_matrix)
true_counts
```

```
##          age          gender          race          genre
##          3           4           4           3
##    energy  danceability  happiness  length_sec
```

```
##           5           3           5           4
##      race:genre      race:energy  race:danceability  gender:danceability
##           2           1           2           2
##      gender:happiness      loudness_db      bpm      age:danceability
##           2           1           1           1
##      race:happiness      race:bpm      race:length_sec      gender:bpm
##           1           1           1           1
```

```
rownames(presence_matrix)[true_counts == max(true_counts)]
```

```
## [1] "energy"      "happiness"
```

```
false_counts <- rowSums(!presence_matrix)
false_counts
```

```
##           age           gender           race           genre
##           3           2           2           3
##      energy      danceability      happiness      length_sec
##           1           3           1           2
##      race:genre      race:energy  race:danceability  gender:danceability
##           4           5           4           4
##      gender:happiness      loudness_db      bpm      age:danceability
##           4           5           5           5
##      race:happiness      race:bpm      race:length_sec      gender:bpm
##           5           5           5           5
```

```
rownames(presence_matrix)[false_counts == max(false_counts)]
```

```
## [1] "race:energy"      "loudness_db"      "bpm"      "age:danceability"
## [5] "race:happiness"  "race:bpm"      "race:length_sec"  "gender:bpm"
```

```
summary_table <- data.frame(
  TRUE_count = rowSums(presence_matrix),
  FALSE_count = rowSums(!presence_matrix)
)
```

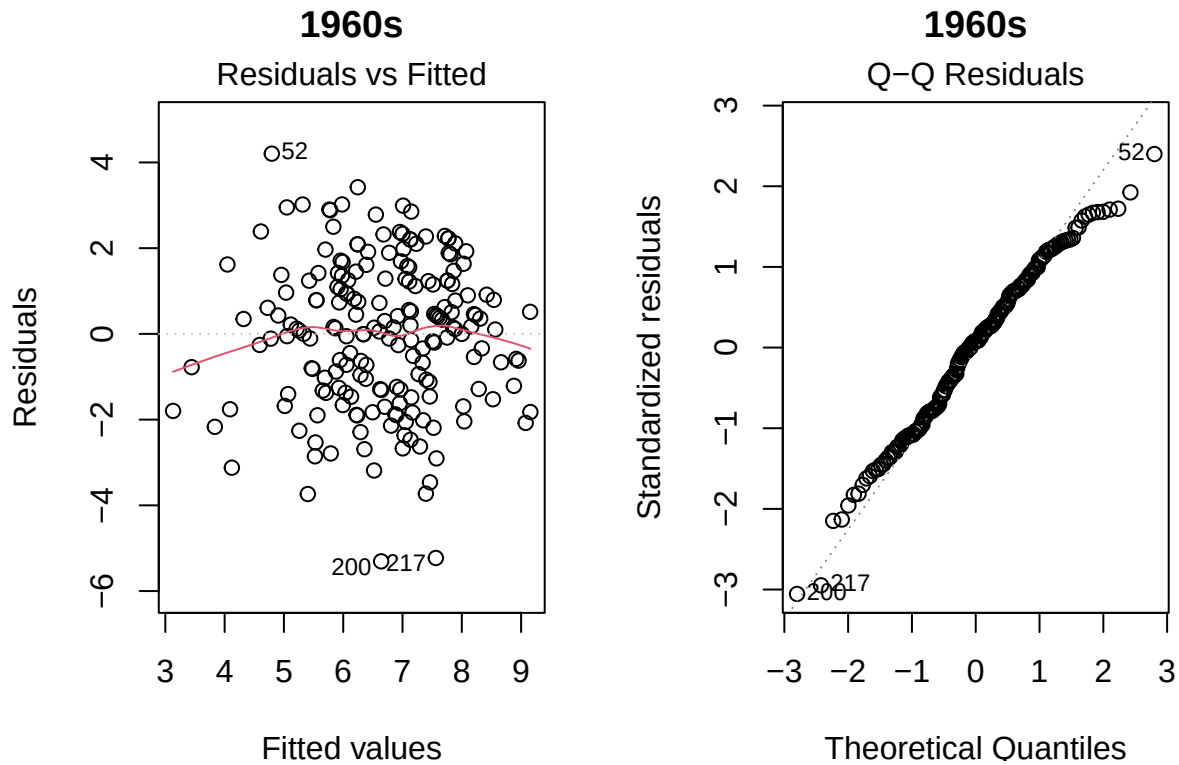
```
summary_table[order(-summary_table$TRUE_count), ]
```

```
##           TRUE_count  FALSE_count
## energy              5             1
## happiness           5             1
## gender              4             2
## race                4             2
## length_sec          4             2
## age                 3             3
## genre               3             3
## danceability        3             3
## race:genre          2             4
## race:danceability   2             4
## gender:danceability  2             4
## gender:happiness    2             4
## race:energy         1             5
## loudness_db         1             5
## bpm                 1             5
## age:danceability    1             5
## race:happiness      1             5
## race:bpm            1             5
```

```
## race:length_sec      1      5
## gender:bpm           1      5

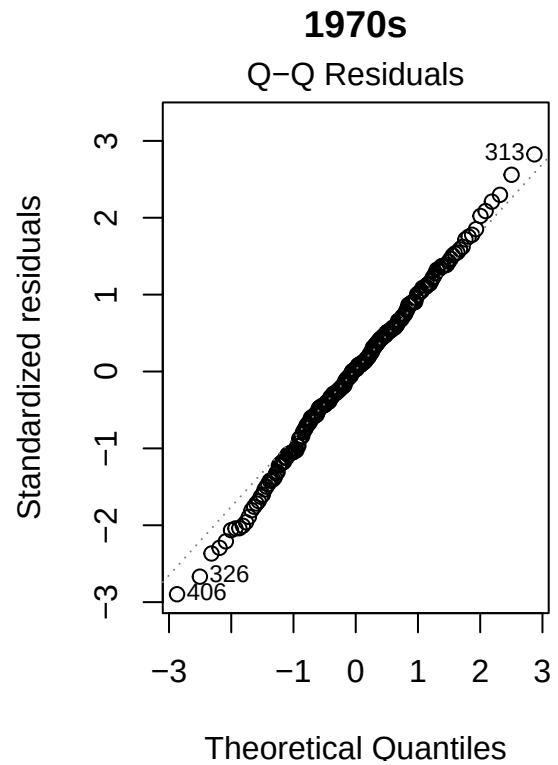
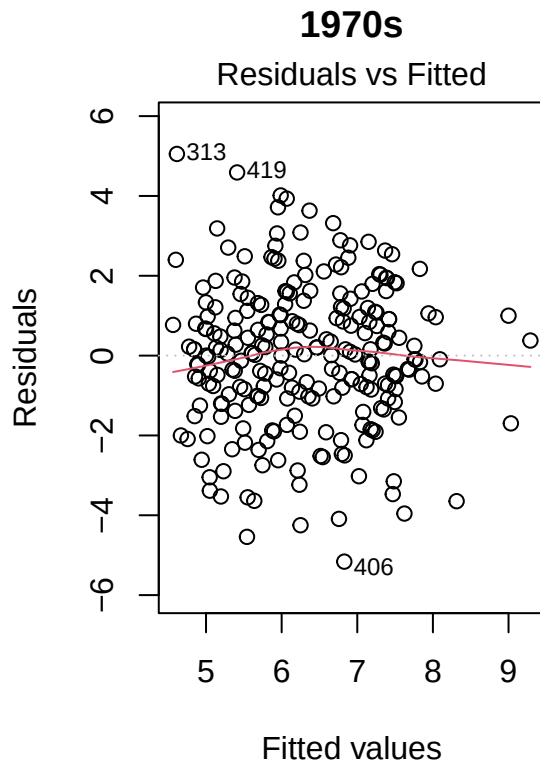
par(mfrow = c(1,2))
plot(model_60s, which = 1:2, main = "1960s")
```

```
## Warning: not plotting observations with leverage one:
##      48
```

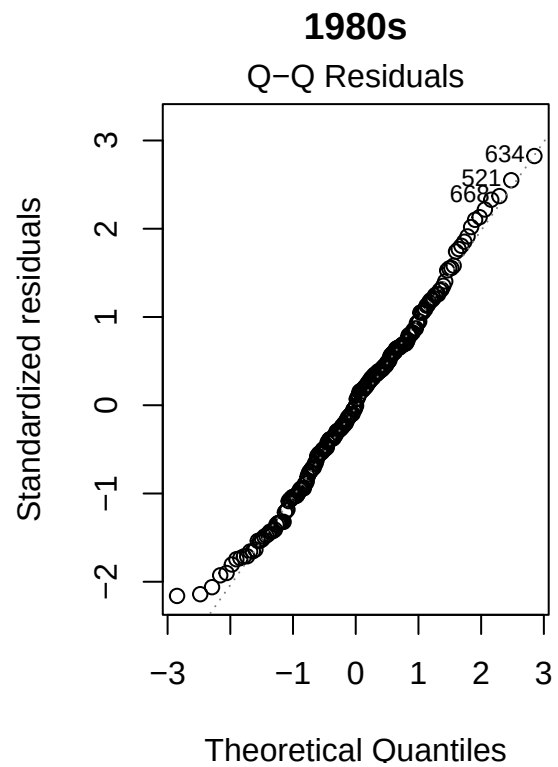
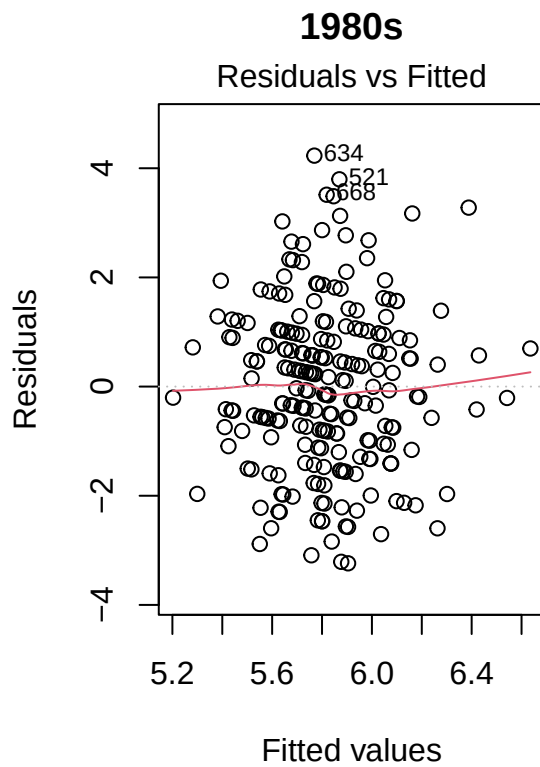


```
plot(model_70s, which = 1:2, main = "1970s")
```

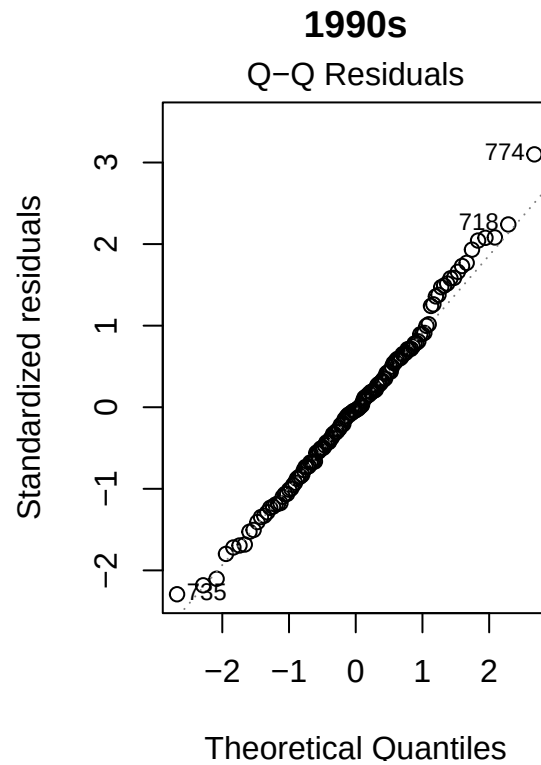
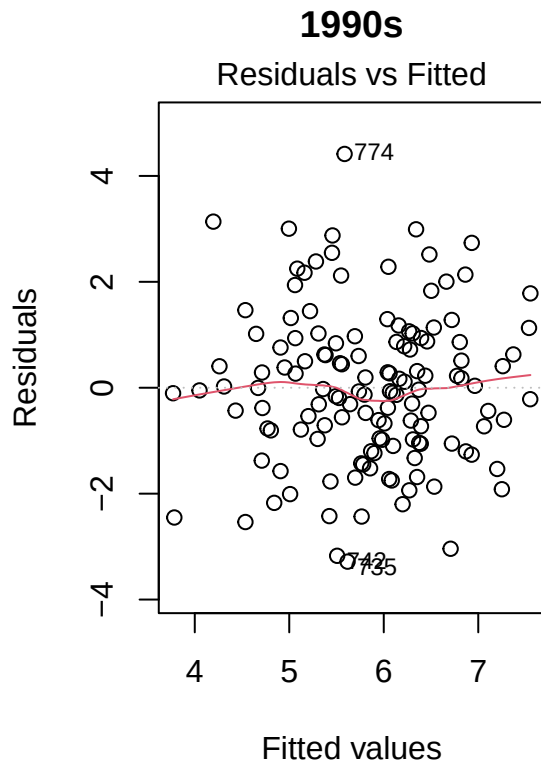
```
## Warning: not plotting observations with leverage one:
##      239
```



```
plot(model_80s, which = 1:2, main = "1980s")
```

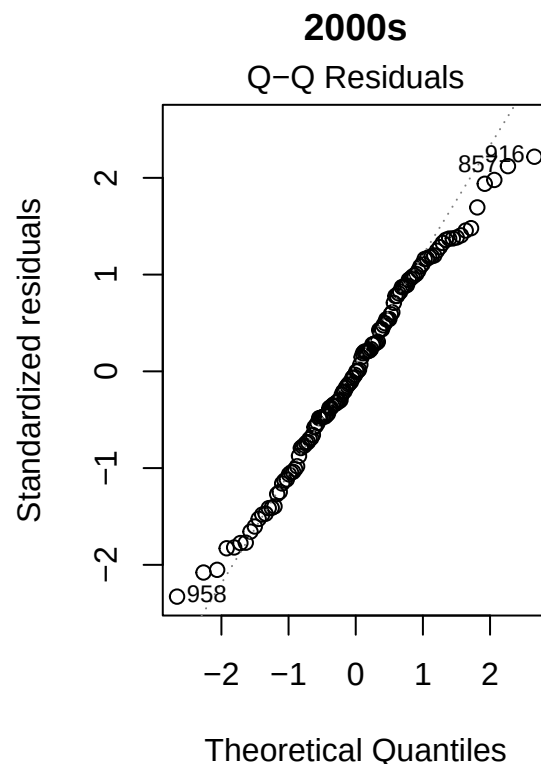
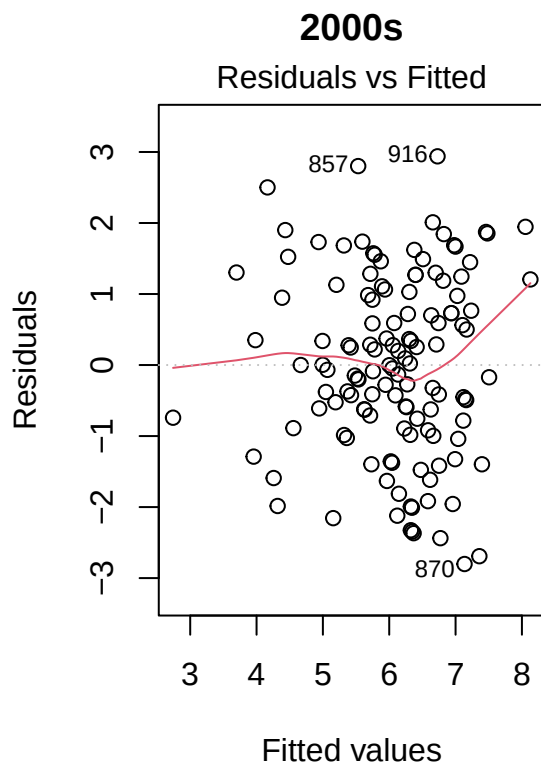


```
plot(model_90s, which = 1:2, main = "1990s")
```

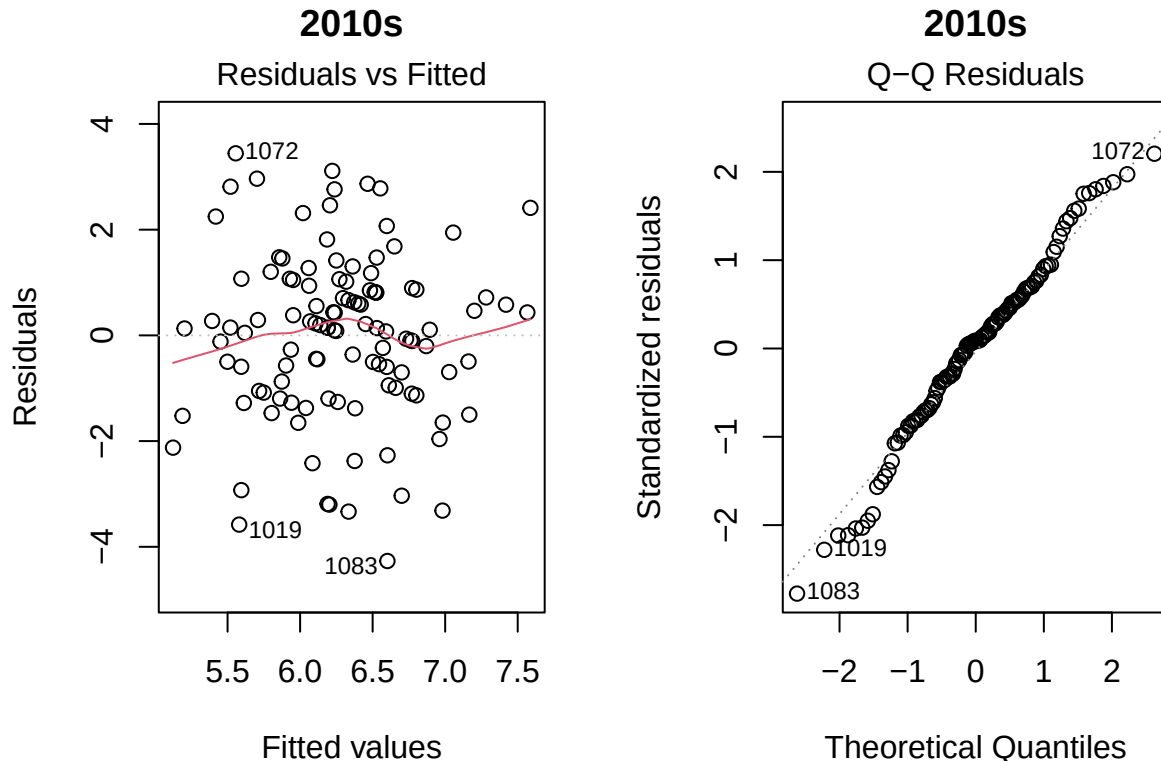


```
plot(model_2000s, which = 1:2, main = "2000s")
```

```
## Warning: not plotting observations with leverage one:
## 4
```



```
plot(model_2010s, which = 1:2, main = "2010s")
```



The fitted regression lines indicate that the relationship between song length and rating varies by genre. In particular, certain genres (e.g., Other and Country) exhibit steeper positive slopes, while others (e.g., Rock and Pop) show weaker or slightly negative trends, suggesting a potential interaction between length and genre.

```
library(broom)
```

```
## Warning: package 'broom' was built under R version 4.4.3
```

```
coef_1960s <- tidy(model_60s, conf.int = TRUE) %>% mutate(decade = "1960s")
coef_1970s <- tidy(model_70s, conf.int = TRUE) %>% mutate(decade = "1970s")
coef_1980s <- tidy(model_80s, conf.int = TRUE) %>% mutate(decade = "1980s")
coef_1990s <- tidy(model_90s, conf.int = TRUE) %>% mutate(decade = "1990s")
coef_2000s <- tidy(model_2000s, conf.int = TRUE) %>% mutate(decade = "2000s")
coef_2010s <- tidy(model_2010s, conf.int = TRUE) %>% mutate(decade = "2010s")

#Appending Data not Represented
#1960s
coef_1960s <- rbind(coef_1960s, list("genreCountry", 0, 0, 0, 0, 0, 0, "1960s"))

#1970s
coef_1970s <- rbind(coef_1970s, list("genreCountry", 0, 0, 0, 0, 0, 0, "1970s"))

#1980s
coef_1980s <- rbind(coef_1980s, list("genreCountry", 0, 0, 0, 0, 0, 0, "1980s"))
coef_1980s <- rbind(coef_1980s, list("genreOther", 0, 0, 0, 0, 0, 0, "1980s"))
coef_1980s <- rbind(coef_1980s, list("genreJazz", 0, 0, 0, 0, 0, 0, "1980s"))
coef_1980s <- rbind(coef_1980s, list("genreRock", 0, 0, 0, 0, 0, 0, "1980s"))
coef_1980s <- rbind(coef_1980s, list("genrePop", 0, 0, 0, 0, 0, 0, "1980s"))
```

```

#1990s
coef_1990s <- rbind(coef_1990s, list("genreCountry", 0, 0, 0, 0, 0, 0, "1990s"))
coef_1990s <- rbind(coef_1990s, list("genreOther", 0, 0, 0, 0, 0, 0, "1990s"))
coef_1990s <- rbind(coef_1990s, list("genreJazz", 0, 0, 0, 0, 0, 0, "1990s"))
coef_1990s <- rbind(coef_1990s, list("genreRock", 0, 0, 0, 0, 0, 0, "1990s"))
coef_1990s <- rbind(coef_1990s, list("genrePop", 0, 0, 0, 0, 0, 0, "1990s"))

#2000s
coef_2000s <- rbind(coef_2000s, list("genreCountry", 0, 0, 0, 0, 0, 0, "2000s"))
coef_2000s <- rbind(coef_2000s, list("genreJazz", 0, 0, 0, 0, 0, 0, "2000s"))

#2010s
coef_2010s <- rbind(coef_2010s, list("genreCountry", 0, 0, 0, 0, 0, 0, "2010s"))
coef_2010s <- rbind(coef_2010s, list("genreOther", 0, 0, 0, 0, 0, 0, "2010s"))
coef_2010s <- rbind(coef_2010s, list("genreJazz", 0, 0, 0, 0, 0, 0, "2010s"))
coef_2010s <- rbind(coef_2010s, list("genreRock", 0, 0, 0, 0, 0, 0, "2010s"))
coef_2010s <- rbind(coef_2010s, list("genrePop", 0, 0, 0, 0, 0, 0, "2010s"))

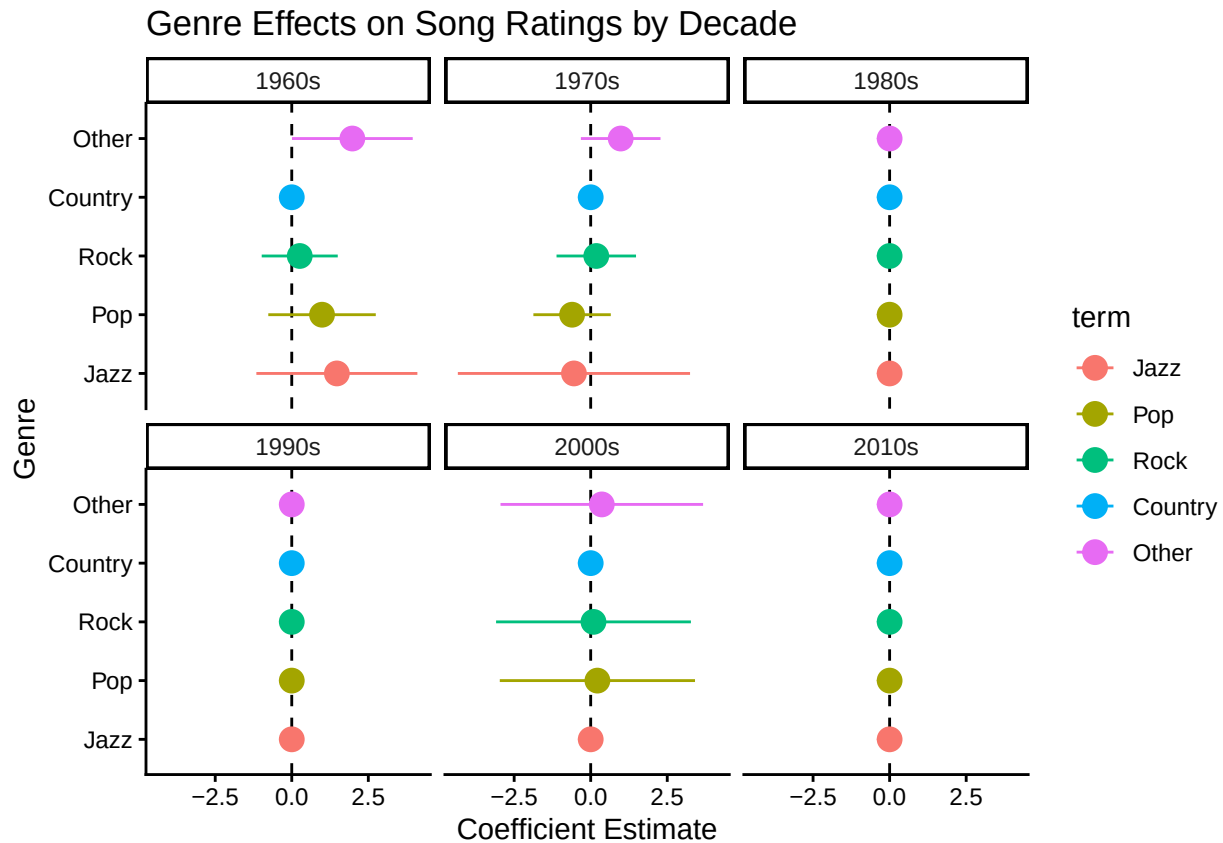
coef_data <- bind_rows(
  coef_1960s,
  coef_1970s,
  coef_1980s,
  coef_1990s,
  coef_2000s,
  coef_2010s
)

coef_data <- coef_data %>%
  filter(term != "(Intercept)") %>%
  filter(grepl("^genre", term)) %>%
  mutate(term = gsub("^genre", "", term))

coef_data$term <- factor(
  coef_data$term,
  levels = c("Jazz", "Pop", "Rock", "Country", "HipHop", "Other")
)

ggplot(coef_data, aes(x = estimate, y = term, color = term)) +
  geom_vline(xintercept = 0, linetype = "dashed") +
  geom_pointrange(aes(xmin = conf.low, xmax = conf.high), size = 0.8) +
  facet_wrap(~ decade) +
  theme_classic() +
  labs(
    title = "Genre Effects on Song Ratings by Decade",
    x = "Coefficient Estimate",
    y = "Genre"
  )

```



model_60s

```
##
## Call:
## lm(formula = rating ~ age + gender + race + genre + energy +
##     danceability + happiness + length_sec + race:genre + race:energy +
##     race:danceability + gender:danceability + gender:happiness,
##     data = subpop)
##
## Coefficients:
##             (Intercept)                  age
##             6.235892                -0.036813
##   genderNot Male          raceWhite
##          -1.820687                1.604179
##   genreJazz             genreOther
##          1.473039                1.978211
##   genrePop             genreRock
##          0.988531                0.260014
##          energy          danceability
##         -0.005765             -0.013030
##          happiness          length_sec
##          0.004845                0.008965
##   raceWhite:genreJazz   raceWhite:genreOther
##          -1.353373             -3.249204
##   raceWhite:genrePop   raceWhite:genreRock
##          -1.786329                      NA
##   raceWhite:energy    raceWhite:danceability
```

```
##              0.024399              -0.046982
## genderNot Male:danceability    genderNot Male:happiness
##              0.076584              -0.030694
```

model_70s

```
##
## Call:
## lm(formula = rating ~ age + race + genre + length_sec, data = subpop)
##
## Coefficients:
## (Intercept)          age    raceWhite    genreJazz    genreOther    genrePop
##   5.250133   -0.033100   -0.430183   -0.547224    0.979427   -0.607790
##  genreRock    length_sec
##   0.182210    0.009494
```

model_80s

```
##
## Call:
## lm(formula = rating ~ energy + danceability + happiness, data = subpop)
##
## Coefficients:
## (Intercept)          energy    danceability    happiness
##   4.960976    0.009412    0.015115   -0.011560
```

model_90s

```
##
## Call:
## lm(formula = rating ~ race + gender + energy + happiness + loudness_db,
##     data = subpop)
##
## Coefficients:
## (Intercept)    raceWhite    genderNot Male          energy    happiness
##   9.46577    -0.46920    0.72094    -0.03512    0.01694
##  loudness_db
##   0.31154
```

model_2000s

```
##
## Call:
## lm(formula = rating ~ age + gender + race + genre + energy +
##     danceability + happiness + bpm + length_sec + age:danceability +
##     race:genre + race:danceability + race:happiness + race:bpm +
##     race:length_sec + gender:danceability + gender:happiness +
##     gender:bpm, data = subpop)
##
## Coefficients:
## (Intercept)          age
##   14.254799   -0.262392
##  genderNot Male    raceWhite
##   -0.551642   -13.479191
##    genreOther    genrePop
##    0.362481    0.218854
##   genreRock          energy
```

```
##           0.090807           0.018882
##           danceability           happiness
##           -0.083852           0.001354
##           bpm           length_sec
##           -0.014549           -0.009048
##           age:danceability           raceWhite:genreOther
##           0.003737           -1.186109
##           raceWhite:genrePop           raceWhite:genreRock
##           NA           NA
##           raceWhite:danceability           raceWhite:happiness
##           0.084078           -0.025028
##           raceWhite:bpm           raceWhite:length_sec
##           0.035295           0.019703
## genderNot Male:danceability           genderNot Male:happiness
##           -0.041970           0.021226
##           genderNot Male:bpm
##           0.019019
```

```
model_2010s
```

```
##
## Call:
## lm(formula = rating ~ gender + energy + happiness + length_sec,
##     data = subpop)
##
## Coefficients:
## (Intercept)  genderNot Male      energy      happiness      length_sec
##    4.788777      0.459667    -0.024064     0.012193     0.009668
```

```
anova(model_results[["1960s"]] $\$$ first, model_results[["1960s"]] $\$$ second)
```

```
## Analysis of Variance Table
##
## Model 1: rating ~ age + race + gender + genre + energy + danceability +
## happiness + bpm + loudness_db + length_sec
## Model 2: rating ~ age + gender + race + age * genre + age * energy + age *
## danceability + age * happiness + age * bpm + age * loudness_db +
## age * length_sec + race * genre + race * energy + race *
## danceability + race * happiness + race * bpm + race * loudness_db +
## race * length_sec + gender * genre + gender * energy + gender *
## danceability + gender * happiness + gender * bpm + gender *
## loudness_db + gender * length_sec
## Res.Df    RSS Df Sum of Sq    F Pr(>F)
## 1      183 673.25
## 2      155 535.98 28    137.27 1.4178 0.09454 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
anova(model_results[["1970s"]] $\$$ first, model_results[["1970s"]] $\$$ second)
```

```
## Analysis of Variance Table
##
## Model 1: rating ~ age + race + gender + genre + energy + danceability +
## happiness + bpm + loudness_db + length_sec
## Model 2: rating ~ age + gender + race + age * genre + age * energy + age *
## danceability + age * happiness + age * bpm + age * loudness_db +
```

```

##      age * length_sec + race * genre + race * energy + race *
##      danceability + race * happiness + race * bpm + race * loudness_db +
##      race * length_sec + gender * genre + gender * energy + gender *
##      danceability + gender * happiness + gender * bpm + gender *
##      loudness_db + gender * length_sec
##      Res.Df      RSS Df Sum of Sq      F Pr(>F)
## 1      231 764.63
## 2      204 666.81 27      97.812 1.1083 0.3327
anova(model_results[["1980s"]] $\$$ first, model_results[["1980s"]] $\$$ second)

## Analysis of Variance Table
##
## Model 1: rating ~ age + race + gender + genre + energy + danceability +
##      happiness + bpm + loudness_db + length_sec
## Model 2: rating ~ age + gender + race + age * genre + age * energy + age *
##      danceability + age * happiness + age * bpm + age * loudness_db +
##      age * length_sec + race * genre + race * energy + race *
##      danceability + race * happiness + race * bpm + race * loudness_db +
##      race * length_sec + gender * genre + gender * energy + gender *
##      danceability + gender * happiness + gender * bpm + gender *
##      loudness_db + gender * length_sec
##      Res.Df      RSS Df Sum of Sq      F Pr(>F)
## 1      216 494.88
## 2      192 459.91 24      34.968 0.6083 0.9247
anova(model_results[["1990s"]] $\$$ first, model_results[["1990s"]] $\$$ second)

## Analysis of Variance Table
##
## Model 1: rating ~ age + race + gender + genre + energy + danceability +
##      happiness + bpm + loudness_db + length_sec
## Model 2: rating ~ age + gender + race + age * genre + age * energy + age *
##      danceability + age * happiness + age * bpm + age * loudness_db +
##      age * length_sec + race * genre + race * energy + race *
##      danceability + race * happiness + race * bpm + race * loudness_db +
##      race * length_sec + gender * genre + gender * energy + gender *
##      danceability + gender * happiness + gender * bpm + gender *
##      loudness_db + gender * length_sec
##      Res.Df      RSS Df Sum of Sq      F Pr(>F)
## 1      122 269.95
## 2       98 217.50 24      52.448 0.9846 0.493
anova(model_results[["2000s"]] $\$$ first, model_2000s)

## Analysis of Variance Table
##
## Model 1: rating ~ age + race + gender + genre + energy + danceability +
##      happiness + bpm + loudness_db + length_sec
## Model 2: rating ~ age + gender + race + genre + energy + danceability +
##      happiness + bpm + length_sec + age:danceability + race:genre +
##      race:danceability + race:happiness + race:bpm + race:length_sec +
##      gender:danceability + gender:happiness + gender:bpm
##      Res.Df      RSS Df Sum of Sq      F      Pr(>F)
## 1      116 270.33
## 2      108 208.61  8      61.719 3.994 0.0003515 ***

```

```

## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

anova(model_2000s, model_results[["2000s"]])$second)

## Analysis of Variance Table
##
## Model 1: rating ~ age + gender + race + genre + energy + danceability +
##   happiness + bpm + length_sec + age:danceability + race:genre +
##   race:danceability + race:happiness + race:bpm + race:length_sec +
##   gender:danceability + gender:happiness + gender:bpm
## Model 2: rating ~ age + gender + race + age * genre + age * energy + age *
##   danceability + age * happiness + age * bpm + age * loudness_db +
##   age * length_sec + race * genre + race * energy + race *
##   danceability + race * happiness + race * bpm + race * loudness_db +
##   race * length_sec + gender * genre + gender * energy + gender *
##   danceability + gender * happiness + gender * bpm + gender *
##   loudness_db + gender * length_sec
##   Res.Df    RSS Df Sum of Sq    F Pr(>F)
## 1      108 208.61
## 2       93 194.20 15    14.415 0.4602 0.9544

anova(model_results[["2010s"]])$first, model_2010s)

## Analysis of Variance Table
##
## Model 1: rating ~ age + race + gender + genre + energy + danceability +
##   happiness + bpm + loudness_db + length_sec
## Model 2: rating ~ gender + energy + happiness + length_sec
##   Res.Df    RSS Df Sum of Sq    F Pr(>F)
## 1       103 274.94
## 2       110 279.47 -7    -4.5351 0.2427 0.9735

anova(model_2010s, model_results[["2010s"]])$second)

## Analysis of Variance Table
##
## Model 1: rating ~ gender + energy + happiness + length_sec
## Model 2: rating ~ age + gender + race + age * genre + age * energy + age *
##   danceability + age * happiness + age * bpm + age * loudness_db +
##   age * length_sec + race * genre + race * energy + race *
##   danceability + race * happiness + race * bpm + race * loudness_db +
##   race * length_sec + gender * genre + gender * energy + gender *
##   danceability + gender * happiness + gender * bpm + gender *
##   loudness_db + gender * length_sec
##   Res.Df    RSS Df Sum of Sq    F Pr(>F)
## 1       110 279.47
## 2        81 206.34 29    73.132 0.9899 0.4938

```